

► How to include styles?

Ans: using <style> tag

style is added using <style> element in the same document

```
<style>
h1 {
    colour: red;
}
</style>
```

► Text properties
Date _____

- 1. text align
- 2. font weight (bold, light, 100-900)
- 3. text decoration
- 4. line height
- 5. letter spacing
- 6. font-size

► Unit (Font size) in CSS

Absolute	Relative
px	% ←
pt	em 1 inch = 96px
pc	rem ↓ pixel
cm	ch
mm	vh
in	vwt many more

► How to include styles?

• External style sheet

↓
Writing CSS in a separate document & linking it with HTML file

```
<head>
<link rel="stylesheet"
href="style.css">
</head>
```

► font-family

↓
Specifies a prioritized list of one or more font family names
Eg: font-family: arial;

► Universal Selector

Syntax: * {
property: value;
}

► Hex codes (base 16)

↓
Hexadecimal code

0-9
A-F

Eg: #ff a5 11
R G B

► ID selector: select an element based on the value of the element's id attribute

Syntax: #myid {
property: value
}

► Class Selector

Syntax: `.myclass {
 property: value;
}`

► descendant selector

Eg: selector that select all paragraphs inside `div`s

Syntax: `div p {
 property: value;
}`

► Adjacent Sibling Combinator

Eg: `h3` that comes immediately after any paragraph

Syntax: `p + h3 {
 property: value;
}`

► Attribute Selector

selects elements based on the presence or value of a given attribute

Eg: `input [attr="value"] {
 property: value;
}`

`h2 {
 background: yellow;
}`

final colour is yellow

► Pseudo class

↓
A keyword added to a selector that specifies a special state of a selected elements

Eg: `element: hover {
 property: value;
}`

`element: active {
 property: value;
}`

`element: checked {
 property: value;
}`

`element: nth-of-type(2) {
 property: value;
}`

► CSS: Cascading style sheet

what is cascade in CSS?

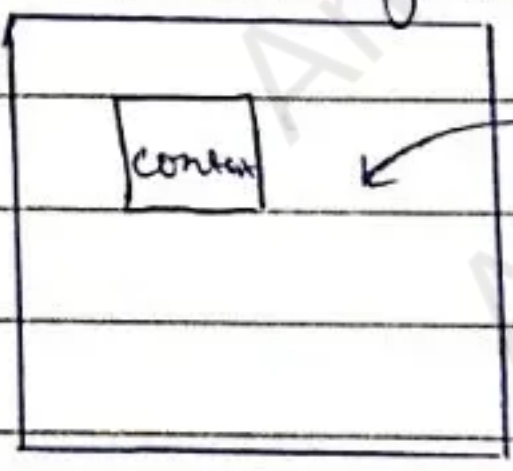
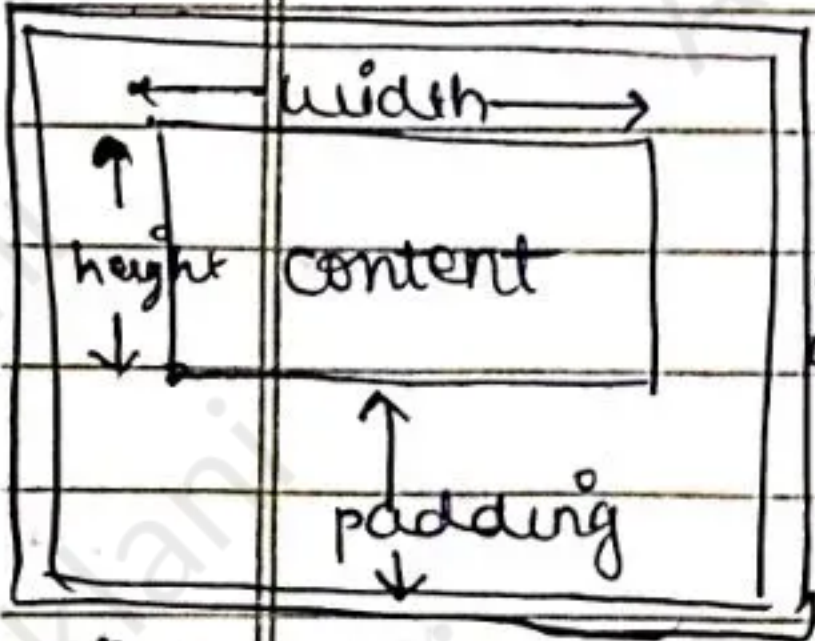
↓
CSS cascade algorithm's job is to select CSS declaration in order to determine the correct values for CSS properties

`h2 {
 background-color: blue;
}`

ε: priority : id > class > element
in terms of selectors

0
id class element

Date

► Important ✓ To show the important element of the doc.	► height element { height: - px }	► Border used to set an element border width style colour border shorthand div { width size color
► Inheritance All elements inherit property excluding input boxes. to inherit the into input input { background property: inherit; } same as in "button"	► Border sides ↓ To control an individual side of the box	border: 2px solid blue;
► NOTE: colour palette are available	► Padding  padding - padding-left - padding-right - padding-bottom - padding-top	► Padding shorthand # all 4 sides padding: 50px # top & bottom left & right padding: 1px 2px # top & left right & bottom padding: 1px 2px 3px
► Property does not inherit, height, width, border	► Margin ↓ To set spacing on the outside of border,	# top right bottom left padding: 1px 2px 3px 4px;
► Box Model in CSS  width height content padding margin border	► Margin shorthand (i) for all sides margin: 50px (ii) top left & right bottom margin: 1px 2px 3px	padding: 1px 2px 3px 4px;

display property
 ↓
 It sets whether an element is treated as a block or inline element and the layout used for its children
 display : inline
 display : block

Block	Inline
• Always start on new line • Take up as much horizontal space as possible (the full width of the container browser window if there is no container) • Will respect width & height CSS properties • Horizontal & vertical margins both work	• Do not start a new line • Only use as much horizontal space as required by the content • Do not accept width & height CSS properties • Margins will work horizontally but not vertically • Padding works on all sides, but the top & bottom may overlap other elements

inline-block (display property)
 ↓
 style - block, behave - inline (single line)

Unit in CSS

Absolute	Relative
px	%
pt	em
pc	rem
cm	ch
mm	vh
in	vw + many more

Percentage (%)
 ↓
 It is often used to define a size as relative to an element's parent object
 Eg : width : 33.33% // relative to the parent
 margin-left : 50% // relative to the parent size

Em
 ↓
 Font-size of the parent, in case of typographical properties like font-size, & font size of the element itself, in the case of other properties like width

Drawbacks of (em)
 ↓
 Snowball effects in nested elements
 To overcome this drawbacks, we have (root em) which set property with root element

alpha channel

↓
is the opacity for a color
ranges from 0 to 1

rgba: (red, green, blue, alpha)

4 a f f 0 0 a 1
r g b alpha

Opacity

↓
sets the opacity for the
element

ranges from 0 to 1
hidden

opacity: 0.5

CSS transitions

transitions enable you to
define the transition between
two states of an element



Syntax: transition: 2s // duration

Transition shorthand

property name | duration | timing-
function | delay

transition: margin-top 2s
ease-in-out 0.2s;

margin: 0 auto;

(i) Rotate (transform
property)
transform: rotate
(45deg); rotate, scale,
skew or translate)

(ii) scale

transform: scale(0.5);

(iii) translate

transform: scale(2, 0.5);

(iv) skew

transform: skew(0.5deg)

Box-shadow

↓ 3D effects
It adds shadow effects
around an element's frame

box-shadow: 2px 2px 10px
green; x-axis y-axis ↓
colour blur

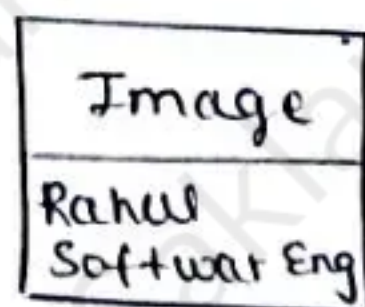
Background Image

↓
lets ~~se~~ you set an image
as a background

background-image: url("path");

background-size: contain / cover /
auto; ↓
 Stretch Cropx Cropx
 Scalex Scalex
 repeat

Card Hover effect



Syntax: card {

width: 200px;

height: 200px;

border: 2px solid black;

display: inline-block

margin: 10px


```
border-radius: 5px;
box-shadow: 2px 4px
          18px black
transition: 2s
```

```
mg {
width: 100%;
border-radius: 5px
5px 0 0;
```

```
card: hover {
box: 0 8px 16px green;
shadow
```

Position

sets how an element is positioned in a document. The "top", "right", "bottom", and "left" properties determine the final location of positioned elements.

- static
- relative
- absolute
- fixed

(i) Position - static

The top, right, bottom, left and z-index properties have no effect. This is a default value.

(ii) Position - relative

The offset is relative to itself based on the values of top, right, bottom & left.

(iii) Position - Absolute

The element is removed from the normal document flow, and no space is created for the element in the page layout.

- It is positioned relative to its closest positioned ancestor, if any; it will be placed relative to the initial containing block.

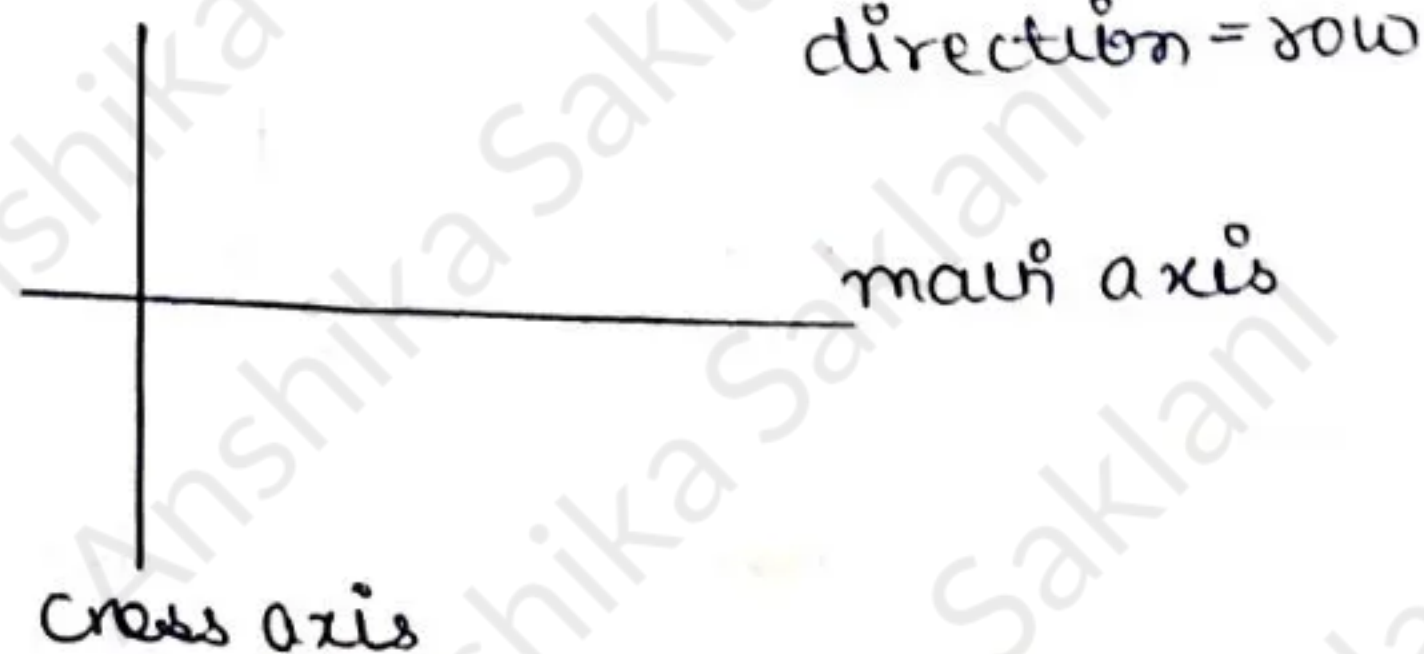
(iv) Position - Fixed

The element is removed from the normal document flow, and no space is created for the element in the page layout.

It is positioned relative to the initial containing block established by the viewport.

Flex box

flexible box layout
make website responsive
arranging items in rows or columns



General example for flex-direction

sets how flex items are placed in the flex container, along which axis and direction

flex-box direction: row; (main axis, L → R)

flexbox-direction: row-reverse; (main axis, right → left)

flex-box direction: column; (main axis, top → bottom)

flexbox-direction: column-reverse; (bottom to top)

Justify Content

tells how the browser distribute space b/w and around content items along main axis

- justify-content: flex-start;
- justify-content: flex-end;
- justify-content: flex-center;

when spacing required

space-between

space-around

space-around

flex-wrap (sets whether flex items are forced onto one line)
flex-wrap: wrap-reverse;

"

"

no wrap
wrap

Align Items

distributes our items along cross axis.

Align items, ~~can align~~ content and Align self

aligns an items along cross axis

flex-sizing

flex-basis

sets the initial main size of a flex item.

flex-basis: 100px;

flex-grow: 3

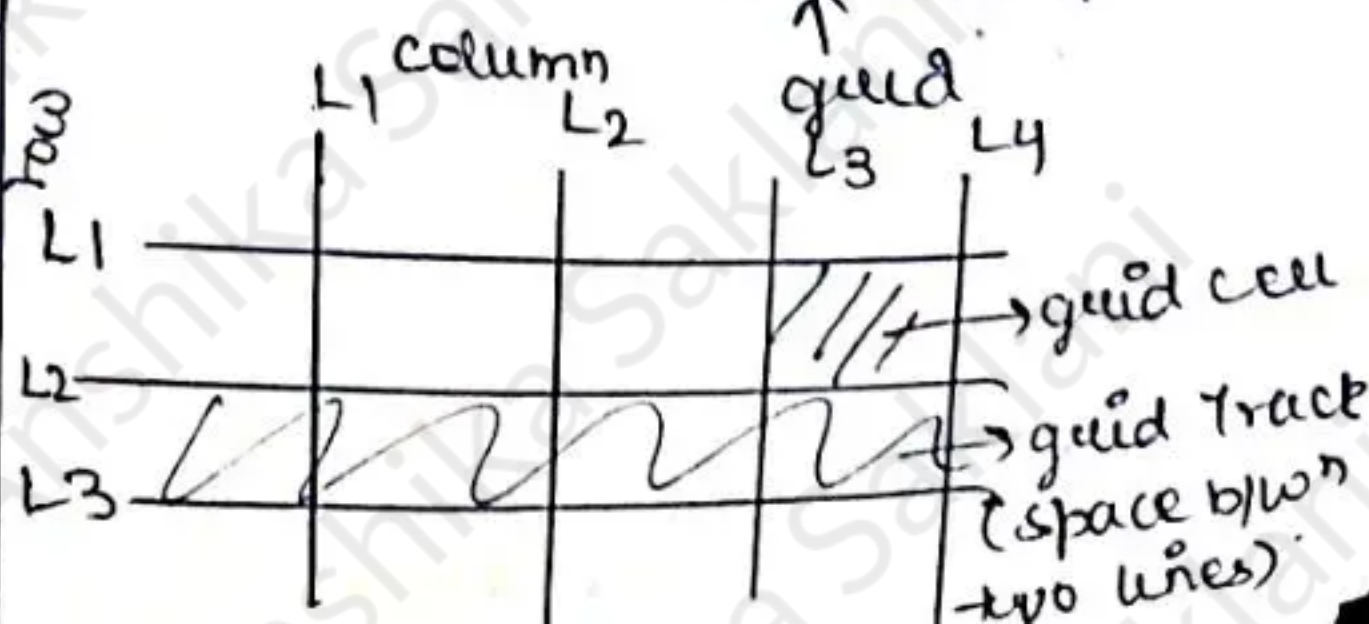
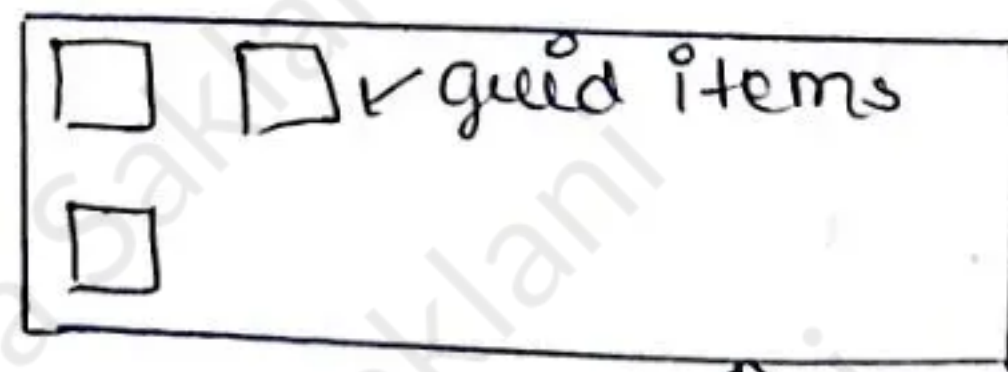
max-width: 400px

min-width: 200px

Grid

setting a container's display to grid will make all children grid items container

display: grid;



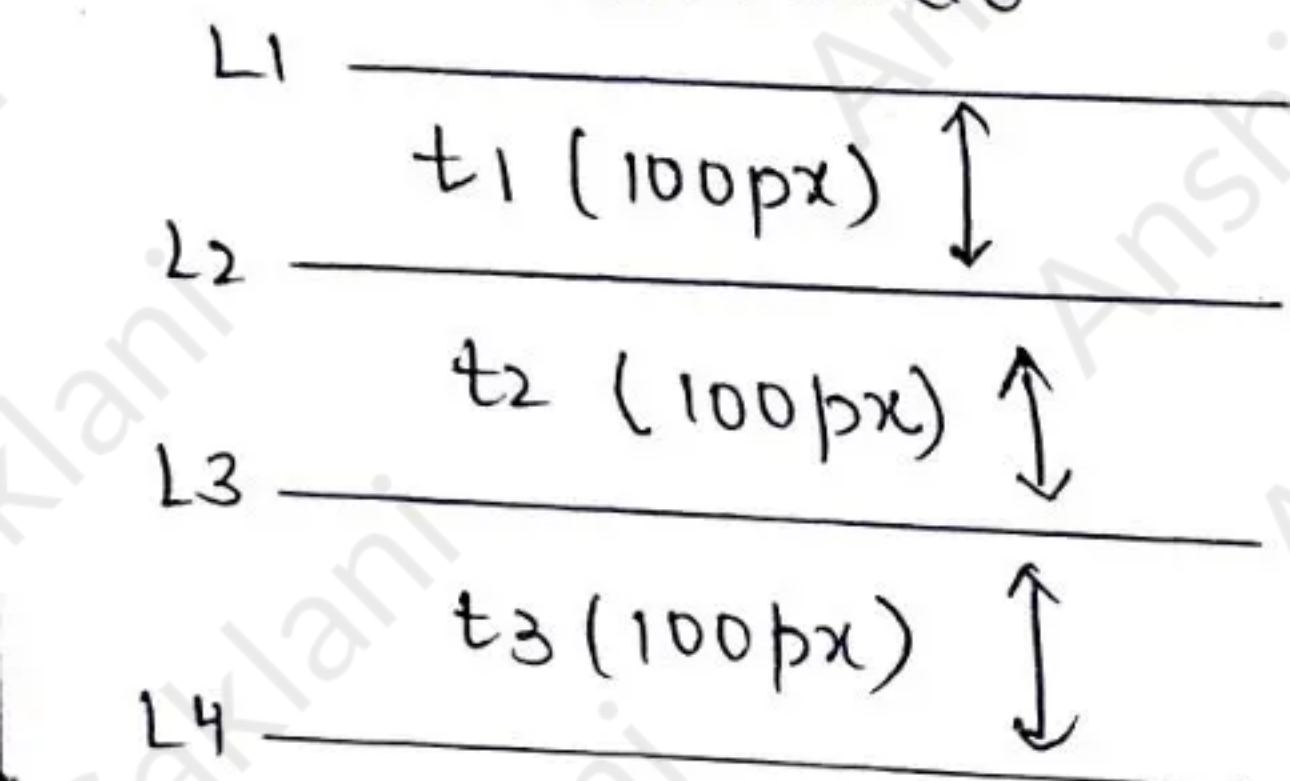
Grid Template

they define the lines & track sizing

Syntax:

grid-template-rows: 100px
100px 100px

grid-template-columns: auto auto auto



"repeat" is used to divide all the available space

Syntax:

grid-template-rows: repeat(5, 1fr);

grid-template-columns: repeat(3, 1fr);

Grid Gape

they define the gaps between the lines

row-gap: 10px

column-gap: 10px

Grid Columns

defines an item's starting & ending position inside the column

Syntax:

grid-column-start: line-number
grid-column-end: line-number

grid-column: start-col / end-col

grid-column: start-col / span number

Grid Rows: defines an item's starting & ending position

grid-row-start: 1

grid-row-end: 3

grid-row: 1/3

grid-row: 1 / span 3

Common Properties

justify-items (container)
justify-self (item)

horizontal

align-items (container)
align-self (item)

vertical

place-items
place-self

CSS Animations

to animate CSS elements

Syntax:

@keyframe myName {

from { font-size: 20px; }

to { font-size: 40px; }

}

Properties:

1. animation-name: fontAnimation

2. animation-duration: 2s;

3. animation-timing-function: ease-in
 4. animation-delay: 0
 5. animation-iteration-count: 1, 2, 3, ... infinite
 6. animation-direction:
 ↙ ↘
 alternate-re normal alternate
 ↘
 reverse

Animation Shorthand
 ↓
 animation: myName 2s
 linear 3s infinite normal

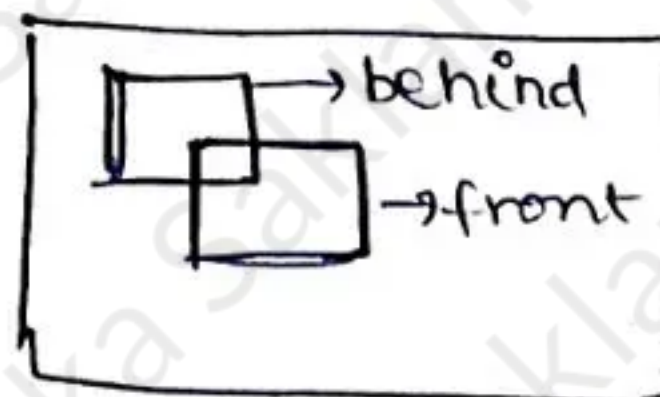
% in Animation
 @keyframe myName {
 0% {font-size: 20px;}
 50% {font-size: 30px;}
 100% {font-size: 40px;}
 }

Media Queries
 ↓
 help create a responsive website
 media features - width
 (of viewport)

device / environment
 ↙ ↘
 width height orientation
 ↓
 landscape / portrait

@media (width: 400px) {
 div {
 min-width max-width
 background-color: red;
 }
 }
 }
Media Queries → orientation
 syntax:
 @media (orientation: landscape) {
 div {
 background-color: red;
 }
 }

z-index
 It decides the stack level of elements
 overlapping elements with a larger z-index cover those with a smaller one
 z-index: auto (0)
 z-index: 1/2/---
 z-index: -1/-2/----



NOTE: No change of z-index will take place if position remains static & default
 Position should be "absolute" or "relative"

► Basic Design Principle

For better looking websites

1. Color
2. Typography

NOTE: Canva.com



for colour wheel theory

- style & appearance of text.
- Font families

► Google Fonts → 1513

↓
different font style

<https://fonts.google.com>

► Bootstrap → understanding

framework/
toolkit for
CSS

flex box
concept

1. pre-built component

► powerful, feature-packed frontend toolkit. Build anything from prototype to production — in minutes

Resources: getbootstrap.com

git init

git add.

git commit -m "first commit"

git branch -M main

git remote add origin "url" git

git push -u origin main

► Resume. Important Tips :

1. ATS (Application Tracking System)
2. Google docs, Canva, resume.com...
3. Main sections

► Features of bootstrap

- Make Development faster & easier
- Good choice of pre-built components
- good browser compatibility

► Link bootstrap with html



CDN (Content delivery network)

► What will we cover?

Components

pre-built and reusable piece of user-interface (UI)

Eg: button, forms

Layouts

Overall structures and arrangements of the elements

Eg: container, grid

► layout

containers are used to contain, pad and (sometimes) center the content within them. Eg: class="container"

► Button

class="button"
Syntax: <button type="button" class="btn"> Button one </button>

► Nav bar

<nav class="navbar bg-body-tertiary fixed-top navbar-expand-lg">

 Brand Name

<div class="navbar-nav">

 Home

 Contact

</div>

</nav>

► layout

• Grid (Responsive) Grid options

	xs (< 576px)	sm	md	lg	xl	xxl
container(max-width)		≥ 576px	≥ 768px	≥ 992	≥ 1200px	≥ 1400px
class prefix	None(auto)	500px	720px	960px	1140px	1320px
# of columns	12					
Gutter width	1.5rem					

► Card

A card is a flexible and extensible content container. It includes options for headers and footers, a wide variety of content, contextual background colours, & powerful display options

► Grid (layout)

uses a series of containers, rows, and columns to layout and align content. It's built with flexbox & is fully responsive.

- 12 column system (12 units)

• Grid (column)

Vertical alignment available

↓
Column